

金融学概论

Time Value of the Money

- ① present value $PV_0 = \frac{FV_n}{(1+i)^n}$ future value at year n
- ② 年金: $PV_0 = \frac{C}{i} \left[1 - \frac{1}{(1+i)^t} \right]$ (每年付 C, 付 t 年); 永续年金 $PV = \frac{C}{i}$
- ③ 年度百分率 APR = period rate $\times$ number of periods per year
- 有效年利率 EFF = $(1 + \frac{APR}{m})^m - 1$
- ④ $1 + \text{名义利率} = (1 + \text{实际利率})(1 + \text{inflation})$

Asset Classes

- ① Discount Yield = $\frac{FV - P}{FV} \times \frac{360}{M}$ Face value price 换为 p 365 or 366 investment yield
- ② $P_B = \sum_{t=1}^T \frac{C_t}{(1+r)^t} + \frac{FV}{(1+r)^T}$ bond price interest / coupon payments 通常: 半年利率 i.e. 每期到期收益率 YTM
- ③ option confers a right to its holder. future 无初始成本, 到期必须交易

	看涨 calls	看跌 puts
买方	有权买	有权卖

- ④ 国债拍卖: 作上者按比例买

Trading, Margin & Short sale

- ① Buying on Margin

Assets	Liabilities & Equity
Shares 10 股票 100P	Loan 4 Equity 6 = 100P - 4 净资产

$$\frac{100P - 4}{100P} = \text{margin} \Rightarrow \text{Margin call}$$

- Short Sale

Assets	Liabilities & Equity
Cash 16	(Shares owed) 10 = 100P Equity 6

$$\frac{16 - 100P}{100P} = \text{margin}$$

Stock Market Indexes

- ① Price weighted Index Value ~

stock split 不影响 index (2-for-1)

Funds

- ① Net Asset Value = $\frac{\text{Market Value} - \text{Liabilities}}{\text{Shares Outstanding}}$

Portfolio Choice (in $E(r)$ - σ space)

- ① Utility function $U = E(r) - \frac{1}{2} A \sigma^2$ 无差异曲线 $E(r_c) = U + \frac{1}{2} A \sigma_c^2$ 比例

- ② 组合 $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2\sigma_{12}$ or $\sum_{i,j} w_iw_j\sigma_{ij}$ 资产配置

- p + f $\Rightarrow$ ③ Sharpe ratio = $\frac{\text{Risk Premium}}{\text{SD of excess return}} = \frac{E(r_p) - r_f}{\sigma_p} \Rightarrow \text{CAL: } E(r_c) = r_f + S \sigma_c$

Index Models

- ① $R_i = \alpha_i + \beta_i R_m + e_i$ var σ_i^2 market index M var σ_m^2 sensitivity to factors that affect all securities

excess return 超额收益率

assume $E(e_i) = 0$ $\text{Cov}(e_i, R_m) = 0$ $\text{cov}(e_i, e_j) = 0$

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i)$$

- ② Diversification:

$$R_p = \sum w_i R_i = \frac{1}{n} \sum \alpha_i + \frac{1}{n} \sum \beta_i R_m + \frac{1}{n} \sum e_i := \alpha_p + \beta_p R_m + e_p$$

$$\sigma_p^2 = \beta_p^2 \sigma_m^2 \quad (\sigma^2(e_p) \rightarrow 0 \text{ as } n \rightarrow \infty)$$

Capital Asset Pricing Model CAPM

- ① 假设投资者均持有市场组合, risk $\uparrow \Rightarrow$ return $\uparrow$

- ② $E(r_i) - r_f = \beta_i (E(r_m) - r_f)$ 超额收益 $\propto$ 敏感度
- $E(r_i) - \beta_i$: Security Market line ($E(r_m)$ 对应 $\beta = 1$)
- $\alpha = R_i - E(R_i)$ average should be 0

Arbitrage Pricing Theory APT

- ① $F := r_m - E(r_m) \Rightarrow r_i = E(r_i) + \beta_i F + e_i$ (well-diversified) $r_p = E(r_p) + \beta_p F$

- ② $\frac{E(r_p) - r_f}{\beta_p} = \frac{E(r_a) - r_f}{\beta_a}$, 否则可构建 portfolio st $\sum \beta_i = 0$

$$E(r_i) - r_f = \beta_i \cdot \lambda$$

Bound Prices and Yields

- ① 半年付息, 但年化利率

$$P/100 = \frac{C/2}{(1+y/2)} + \frac{C/2}{(1+y/2)^2} + \dots + \frac{1+C/2}{(1+y/2)^{2T}}$$

coupon rate 年化票面利率 year to maturity

- ① accrued interest = $C \cdot \frac{N_t}{D}$ 天数为已计息天数

计算: date $\sim$ coupon $\Rightarrow \frac{2.5}{1.25} \Rightarrow P = \frac{2.25}{(1+\frac{y}{2})^{1.25}} + \frac{1}{(1+\frac{y}{2})^{1.25}} + \dots + \frac{102.25}{(1+\frac{y}{2})^{23.75}}$

- ① $DV01 = - \frac{\Delta P}{\Delta y} = 0.01\%$

The Term Structure of Interest Rates

$$(1+f_n) = \frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}}$$

- ② 流动性溢价: $f_n > E(r_n)$

股利 Equity Valuation

① $E(r) = \frac{E(D_1) + E(P_1) - P_0}{P_0}$ ← 买入价

② 市场资本化率/必要收益率 $k = r_f + \beta(E(r_m) - r_f)$

③ fundamental / intrinsic value $k = E(r) = \frac{E(D_1 + EP_1 - V_0)}{V_0}$ → get V_0

④ 股息贴现模型 DDM: 固定增长 rate g , $V_0 = \frac{P_1}{k-g}$, multi-stage 则分开算 $\sum_{i=1}^n \frac{D_i}{(1+k)^i} + \frac{D_{n+1}}{(k-g)(1+k)^{n+1}}$

⑤ price-earnings ratios $D_1 = (1-b)E_1$, $g = ROE \times b \Rightarrow P_0' = \frac{E_1(1-b)}{k-g}$

⑥ $P_0' = \frac{E_1}{k} + PVGO$ → if $ROE > k$

⑦ $\frac{P_0}{E_1} = \frac{1-b}{k-g}$: 股利 $\times \frac{P}{E}$ = 当前股价

Option Strategies & Put-Call Parity

① Covered call: 买股票 + 卖看涨期权

protective put: 买股票 + 买看跌期权

② Put-Call Parity Thm: $S_0 + P = C + \frac{X}{(1+r_f)^T}$

Options Pricing

③ Hedge ratio $H = \frac{C^+ - C^-}{S^+ - S^-}$: write 1 option, buy H shares

→ risk-free $\Rightarrow$ must earn risk free rate

$V = HS^+ - C^+ = HS^- - C^-$

$C_0 = HS_0 - \frac{V}{1+r_f \Delta t}$

套利: 做多便宜的, 做空贵的

	当期	到期	
		$S_T \leq X$	$S_T > X$
Buy call	$-C$	0	$S_T - X$
Buy Bills	$-\frac{X}{(1+r_f)^T}$	X	X
Short Stock	S_0	$-S_T$	$-S_T$
Write Put	P	$S_T - X$	0
Total	$\dots$	= 0	

价格范围

no-arbitrage

④ Monitoring costs: CF_f 时耗审计成本 c $(1+r)L = \alpha_s(1+r_L)L + \alpha_f(CF_f - 1)$

Others

① 持有期收益率 $HPR = \frac{\text{卖出} - \text{买入}}{\text{买入}} = \frac{P_1}{P_0} - 1$

② 股指期货: $F_0 = S_0(1+r_f) - D$

③ "p+f": 最优风险资产权重 $w^* = \frac{r_p - r_f}{A\sigma_p^2}$

④ 存款套利: 借入 $\rightarrow$ 换汇 $\rightarrow$ 存贷 $\rightarrow$ 锁汇

⑤ 系统性风险占比: $R^2 = \frac{\beta_i^2 \sigma_m^2}{\sigma_i^2}$: "风险" 指方差

⑥ CAPM $\beta_i = \rho \cdot \frac{\sigma_i}{\sigma_m}$

Future

① $F_0 = S_0(1+r_f - d)^T = \frac{D}{S_0}$

Structured Finance

② 同时违约 $P_{00} = P_0^+ + P P_0(1-P_0)$

Financial Fragility

Asymmetric Information in Financial Market

$EV = \alpha_s(1+r_L)L$

① 对称则 $(1+r)L = \alpha_s(1+r_L)L + (1-\alpha_s)CF_f$ $E\pi_{\text{borrow}} + E\pi_{\text{lender}} = EV$ 项目期望收益

② Adverse selection $1+r = P_A[\alpha_A(1+r_L)] + P_B[\alpha_B(1+r_L)]$

③ Moral Hazard: 需 $E\pi_H \geq E\pi_L$ "安全项目对利率更敏感"

$EV_H - \alpha_H(1+r_L)L \geq EV_L - \alpha_L(1+r_L)L$
 $(1+r_L) < \frac{EV_H - EV_L}{(\alpha_H - \alpha_L)L} = \frac{\alpha_H CF_{H,S} - \alpha_L CF_{L,S}}{(\alpha_H - \alpha_L)L} \rightarrow \text{对比} \begin{cases} \text{所需 min} \\ \text{不切换上限} \end{cases}$
 $\rightarrow \text{debt: } L \uparrow \Rightarrow \text{moral hazard} \uparrow$

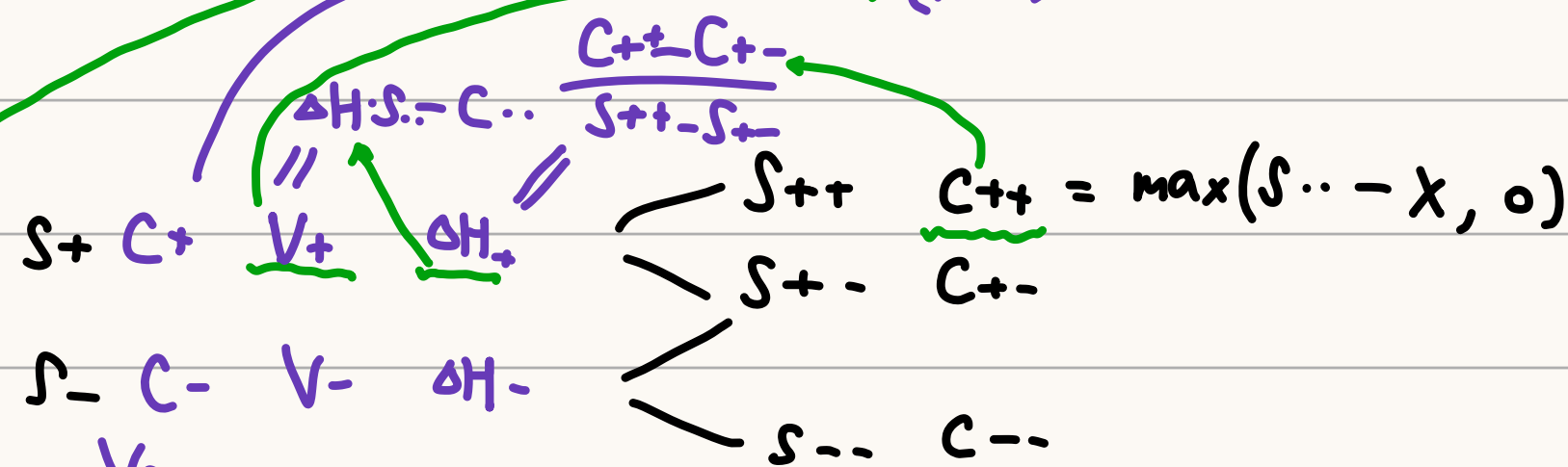
考题总结

$$\Delta H_+ \cdot S_+ - C_+ = \frac{\text{终值}}{(1+r)\Delta t}$$

① 期权 = 叉树

call price

$$\begin{pmatrix} S_0 \\ C_0 \end{pmatrix} = \begin{pmatrix} V_0 \\ \Delta H \cdot S_0 - \frac{V_0}{(1+r)^{\Delta t}} \end{pmatrix}$$



call $\rightarrow$ put: $P_0 = C_0 - S_0 + \frac{X}{1+r_f}$

